B. Customising the Track

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: combinatorics, greedy, math | https://codeforces.com/problemset/problem/1543/B

Highway 201 is the most busy street in Rockport. Traffic cars cause a lot of hindrances to races, especially when there are a lot of them. The track which passes through this highway can be divided into n sub-tracks. You are given an array a where a_i represents the number of traffic cars in the i-th sub-track. You define the inconvenience of the track as $\sum_{i=1}^{n} \sum_{j=i+1}^{n} |a_i - a_j|$, where $|x|$ is the absolute value of x.

You can perform the following operation any (possibly zero) number of times: choose a traffic car and move it from its current sub-track to any other sub-track.

Find the minimum inconvenience you can achieve.

Input

The first line of input contains a single integer t ($1 \le t \le 10\,000$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single line containing a single integer: the minimum inconvenience you can achieve by applying the given operation any (possibly zero) number of times.

Example

Input

```
3
3
1 2 3
4
0 1 1 0
10
8 3 6 11 5 2 1 7 10 4
```

Output

```
0
4
21
```

Note

For the first test case, you can move a car from the 3-rd sub-track to the 1-st sub-track to obtain 0 inconvenience.

For the second test case, moving any car won't decrease the inconvenience of the track.